

AI Operations (AIOps) Assignment 1

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Question 1 — Conceptual: Technical Debt Diagnosis

1.

- **(a) Entanglement (CACE)**

Changing the delivery time feature unexpectedly affected the favorite restaurants feature. This is “Changing Anything Changes Everything.” A small change to one feature unexpectedly affects another part of the model’s behaviour.

- **(b) Undeclared Consumers**

The marketing team was using the model’s output without telling the ML team. This creates a hidden dependency because the ML team doesn’t know who is using the output and that block safe changes.

- **(c) Configuration & Glue-Code Debt**

The training pipeline has 14 undocumented shell scripts and no proper system to manage them.

This makes the pipeline messy, difficult to understand, and difficult to maintain.

2.

Specific mitigation for (c) Configuration & Glue-Code Debt :

Convert the shell scripts into an MLflow Project

1. Instead of 14 separate shell scripts, we create one `MLproject` file in root of repository . This file clearly declares The conda environment , Named entry points instead of loose shell scripts ,Parameters that were previously hard-coded across scripts.
2. Each former shell script becomes a clean, documented entry point.
3. Run the entire pipeline with a single, reproducible command

This solution removes the undocumented scripts, makes the pipeline clear ,Gives a single source of truth for the training workflow and Makes the pipeline version controlled with Git.